AINEBYONA ABUBAKER

Machine Learning and Backend Engineer

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PROFESSIONAL SUMMARY

Machine Learning and backend engineer who builds AI systems and the production platforms around them: model training and optimization, API design, billing, security, deployment, and monitoring. Founder of FaceAttend, a multi-tenant face-verification SaaS with anti-spoofing, subscription billing, and an Enterprise API. Works in Python, FastAPI, PyTorch, PostgreSQL, and Docker on Linux.

SKILLS

Machine Learning: Supervised Learning, Deep Learning, Computer Vision, NLP, Transformers, LLMs, Feature Engineering, Anomaly Detection, CNNs, RNNs, YOLO, Face Recognition
ML Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, InsightFace, OpenCV, ONNX Runtime, Pandas, NumPy
Backend and Infrastructure: Python, FastAPI, PostgreSQL, Supabase, Docker, Nginx, Linux, Datadog, Sentry, Cloudflare, JWT, Row-Level Security
Mobile and Web: Flutter, Android, iOS, HTML, JavaScript

WORK EXPERIENCE

FaceAttend - Founder, ML and Backend Engineer | Aug 2025 - Present
Built and deployed a multi-tenant face-verification attendance SaaS for university use, owning the full lifecycle from concept to production.

- Cut face-verification latency by 97% (1.29s to 0.03s) by optimizing the inference pipeline with InsightFace (ArcFace embeddings) and ONNX Runtime.
- Integrated custom liveness-detection models to block proxy and spoofing attempts.
- Designed an async FastAPI backend across multiple services (main API, MVP API, Nginx reverse proxy), containerized with Docker and self-hosted on a Linux cloud server with SSL via Certbot and DNS/WAF managed through Cloudflare.
- Implemented role-based access control (central admin, department admin, coordinator, lecturer) with row-level security policies across 8+ database tables, JWT authentication, private signed storage URLs, and audit logging with a scheduled daily email digest.
- Shipped a key-authenticated Enterprise API so institutional clients can integrate attendance data into their own systems.
- Built a subscription billing engine: live Mobile Money and card payments via Pesapal, prorated plan changes, automated renewals with APScheduler, PDF invoices with ReportLab, and an internal MRR/ARR/churn analytics dashboard.
- Instrumented production infrastructure with Datadog and Sentry for system, container, network-level, and error monitoring.
- Developed the companion Flutter app (Android and iOS) with Supabase authentication, biometric login, real-time face capture, and Shorebird over-the-air updates, plus an HTML/JavaScript admin dashboard.

PROJECTS

African SMS Spam Classifier | 2026 - Present

- Fine-tuned a DistilBERT transformer on African mobile-money SMS fraud data, reaching 96.7% classification accuracy.
- Deployed as a containerized FastAPI service with single and batch classification endpoints, hosted on the same production infrastructure as FaceAttend.
- Published the fine-tuned model on Hugging Face Hub for public use and reproducibility.

Email Spam Classifier | 2024 - Present

- Fine-tuned DistilBERT on the SMS Spam and Enron datasets for binary text classification, reaching 98% accuracy.
- Deployed with an interactive Gradio interface on Hugging Face Hub.

EDUCATION

Bachelor of Science in Information Systems and Technology | Nkumba University, Wakiso, Eastern, Uganda | Expected 2027

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification - DeepLearning.AI and Stanford University
- Advanced Learning Algorithms; Unsupervised Learning, Recommenders, Reinforcement Learning - DeepLearning.AI and Stanford University
- Neural Networks and Deep Learning; Convolutional Neural Networks; Improving Deep Neural Networks; Structuring Machine Learning Projects - DeepLearning.AI
- Generative AI with Large Language Models - DeepLearning.AI and AWS
- ChatGPT Prompt Engineering for Developers; Machine Learning in Production - DeepLearning.AI
- Generative AI: Prompt Engineering Basics; Python for Data Science, AI and Development; Data Analysis with Python; Databases and SQL for Data Science with Python - IBM
- Python Essentials for MLOps - Duke University
- Networking Fundamentals - Packt